

KENYA HIGH 2025 PREDICTION



KCSE TEST EXAM

121/2

MATHEMATICS

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions to candidates.

- a) Write your name, index number, admission number and name of your school in the spaces provided above
- b) Sign and write the date of the examination in the spaces provided above
- c) This paper consists of two sections: Section I and Section II.
- d) Answer all the questions in Section I and only five questions from Section II
- e) Show all the steps in your calculations, giving your answer at each stage in the spaces provided below each question
- f) Marks may be given for correct working even if the answer is wrong.
- g) Non-programmable silent electronic calculators and KNEC Mathematical tables may be used, except where stated otherwise.

For Examiner's use only.

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand
Total

SECTION 1 (50 MARKS)

Answer all questions in the spaces provided.

1. Use **logarithms** to evaluate.

(4marks)

$$\sqrt{\frac{(0.08294)^2 \times 39.24^3}{8458}}$$

2. Make **x** the subject of the formula.

(3 marks)

$$y = \sqrt{\frac{x}{x+h}}$$

3. Given that $\frac{8}{4-2\sqrt{3}} = a + b\sqrt{3}$ and that **a** and **b** are rational numbers, find the values of **a** and **b**.

(3 marks)

4. Given that $\mathbf{A} = \begin{pmatrix} 1 & 5 \\ 3 & 7 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 7 & 3 \\ -4 & -2 \end{pmatrix}$ and that $\mathbf{C} = \mathbf{AB}$, find $\mathbf{C}^{-1}$ **(3 marks)**

5. (a) Expand and simplify $(1 + 2x)^7$ up to the term in x^3
(2 marks)

b) Use the expansion in (a) above to estimate the value of $(1.02)^7$ correct to four decimal places.

(2 marks)

6. Solve for x in: $\log(7x + 2) - \log(x - 1) = 0$
(3 marks)

7. Find the quartile deviation for the following set of data

(3 marks)

16, 42, 41, 6, 20, 28, 19, 23, 15

8. The radius and height of a cylindrical tank rounded to 1cm are 105cm and 300cm respectively.

Calculate the percentage error in its volume. (to 4 s.f)

(3 marks)

9. Given that $25x^2 - 20x + k$ is a perfect square. Find the value of **k**.

(2

marks)

10. Solve for **X** in the equation:

$$2 \sin^2 x - 1 = \cos^2 x + \sin x, \text{ for } 0^\circ \leq x \leq 360^\circ$$

(3mks)

11. **P** and **Q** are two points on the earth's surface. Their positions are **P** (60°S , 30°E) and **Q** (60°S , 90°W). Find the distance between **P** and **Q** along the parallel of latitude in km (Take radius of earth = 6370 km and $\pi = \frac{22}{7}$) [to 1 decimal place.] **(3 marks)**

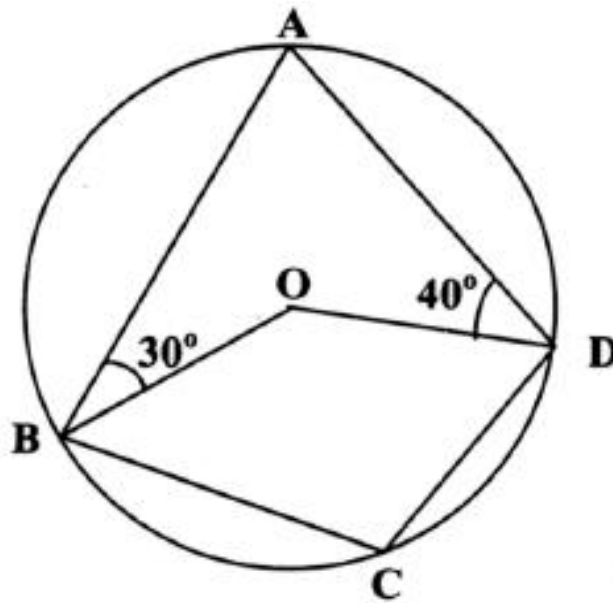
12. a) The sum of the fifth and sixth term of an AP is 30. If the third is 5, find the first term.

(2 marks)

b) The sum of the first 10 terms of the AP

(2 marks)

13. In the figure below, ABCD is a cyclic quadrilateral. Point O is the centre of the circle. Angle ABO = 30° and angle ADO = 40°



Calculate the size of angle BCD.

(2 marks)

14. Obtain the Centre and radius of the circle represented by the equation.

$$x^2 + y^2 - 10y + 16 = 0$$

(3 marks)

- 15.** John bought 3 brands of tea **A, B** and **C**. The cost price of the three brands was sh.25, sh.30 and sh.45 per kilogram respectively. He mixed the three brands in the ratio 5:2:1 respectively. After selling the mixture he made a profit of 20%. How much profit did he make per kilogram of the mixture?

(4mks)

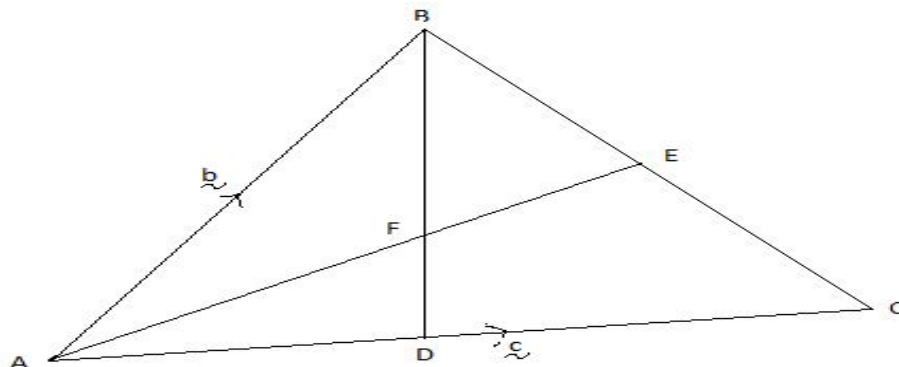
- 16.** Draw a line $DF = 4.6\text{cm}$. Construct the locus of point K above DF such that angle $DKF = 70^\circ$.

(3 marks)

SECTION II (50 MARKS)

(Answer ONLY FIVE questions in the spaces provided)

17. In the figure below, E is the midpoint of BC. $AD : DC = 3:2$ and F is the meeting point of BD and AE.



(a) Express the following vectors in terms of $\mathbf{b}$ and $\mathbf{c}$

(i) $\overrightarrow{BD}$

(2 marks)

(ii) $\overrightarrow{AE}$

(2 marks)

(b) If $\mathbf{BF} = t\mathbf{BD}$ and $\mathbf{AF} = n\mathbf{AE}$, find the value of t and n
marks)

(5

(c) State the ratio in which F divides BD and AE

(1 mark)

18. The probability that a boy goes to school by bus is $\frac{1}{3}$ and by matatu is $\frac{1}{2}$. If he uses a bus the probability that he is late to school is $\frac{1}{5}$ and if he uses a matatu the probability of being late is $\frac{3}{10}$. If he uses other means of transport, the probability of being late is $\frac{1}{20}$.

(a) Represent this information on a tree diagram.

(2 marks)

(b) What is the probability that;

(i) He will be late to school

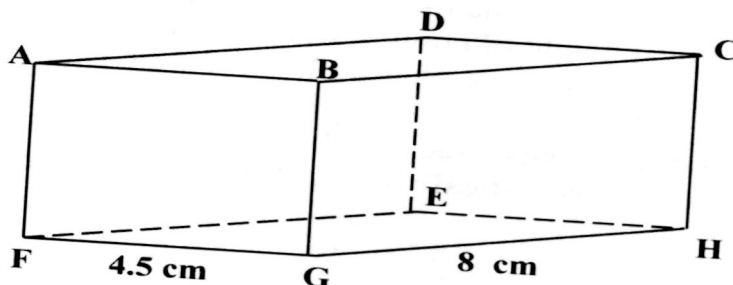
(2 marks)

(ii) He will not be late to **school**
(2 marks)

(iii) He will be late to school if he does not use a matatu. (2 marks)

(iv) He neither uses a bus nor matatu but arrives to school early. (2 marks)

19. The diagram below represents a cuboid ABCDEFGH in which $FG = 4.5$ cm, $GH = 8$ cm and $HC = 6$ cm



Calculate:

- (a) The length of FC (2 mks)
- (b) (i) the size of the angle between the line FC and the plane FGHE (2 mks)

(ii) The size of the angle between the lines AB and FH

(2 mks)

(iii) The size of the angle between the lines AC and GE

(2mks)

(c) The size of the angle between the planes ABHE and the plane FGHE

(2 mks)

20. The table below shows the income tax rates

Income per annum in K£	Rate in Sh per K£
1-1980	2
1981-3960	3
3961-5940	5
5941-7920	7
7921-9920	9
Over 9920	10

Juma is a salaried employee with a basic salary of Ksh. 15,710 per month and house allowance of sh1,350 monthly. He is also entitled to a monthly tax relief of Ksh1320.

Calculate:

(a) Juma's annual salary in K£.

(2 marks)

(b) The monthly tax paid by Juma in Ksh.

(6 marks)

(c) Find his net pay per month in Ksh.

(2 marks)

21. A transformation represented by the matrix $\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$ maps the points A(0, 0), B(2, 0), C(2, 3) and

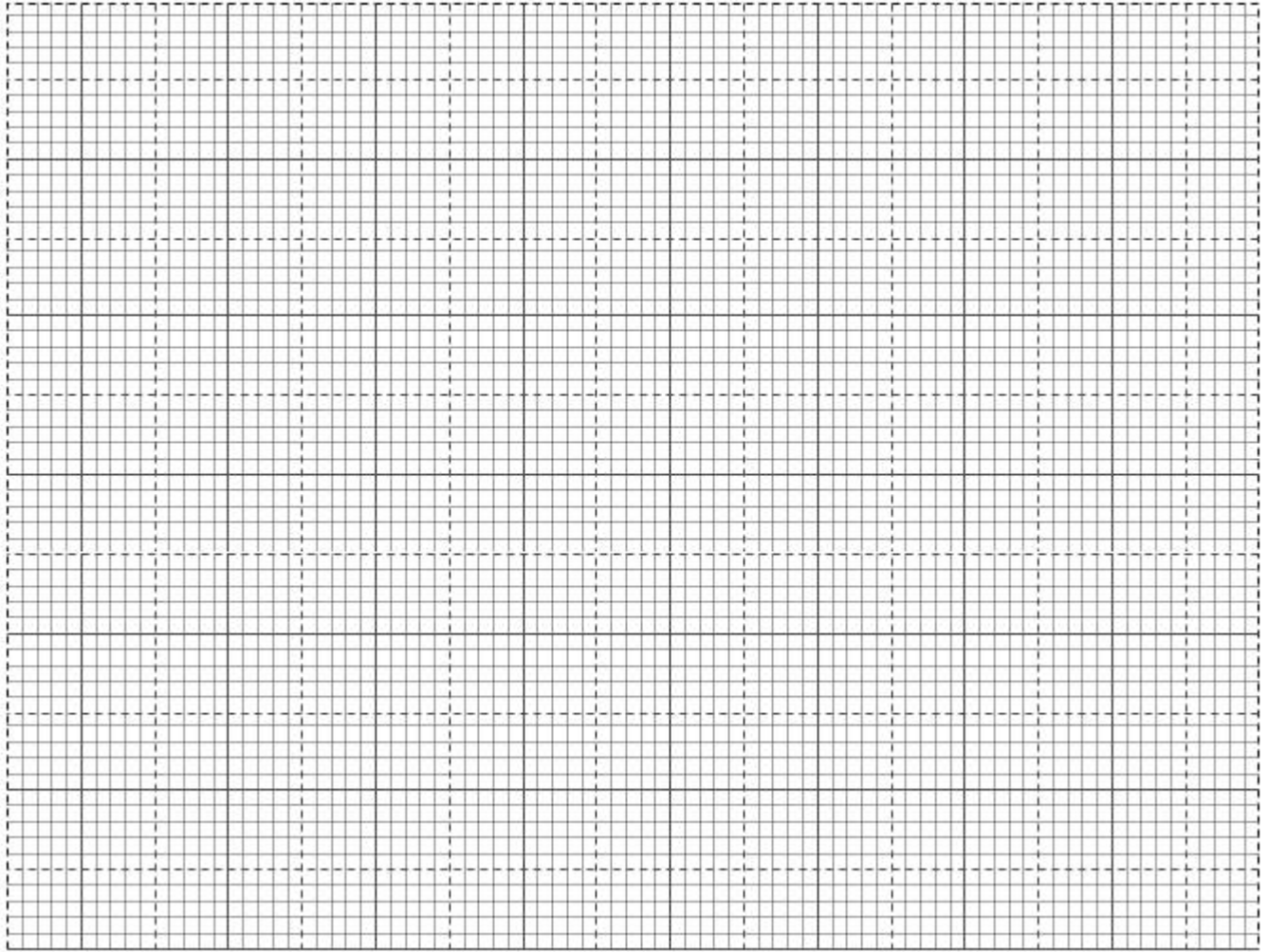
D(0, 3) onto A'B'C'D' respectively.

a. Find the co-ordinates of A'B'C'D'.

(2 mks)

b. Draw the quadrilateral ABCD and its image A'B'C'D'.

(2mks)



c. Hence or otherwise, determine the area of $A^1B^1C^1D^1$. **(2mks)**

d. Another transformation represented by the matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ maps $A^1B^1C^1D^1$ onto $A^{11}B^{11}C^{11}D^{11}$. Draw the image $A^{11}B^{11}C^{11}D^{11}$. **(2mks)**

e. Determine the single matrix, which maps $A^{11}B^{11}C^{11}D^{11}$ back to ABCD. **(2mks)**

22. A particle starts from rest at a point A and moves along a straight line coming to rest at another point

B. During the motion, its velocity $v(\text{m/s})$ after time t (sec) is given by $v=9t^2-2t^3$. Calculate:

a) The time taken for the particle to reach B. **(2 marks)**

b) The distance traveled during the first two seconds. **(3 marks)**

c) The time taken for the particle to attain its maximum velocity. **(3 marks)**

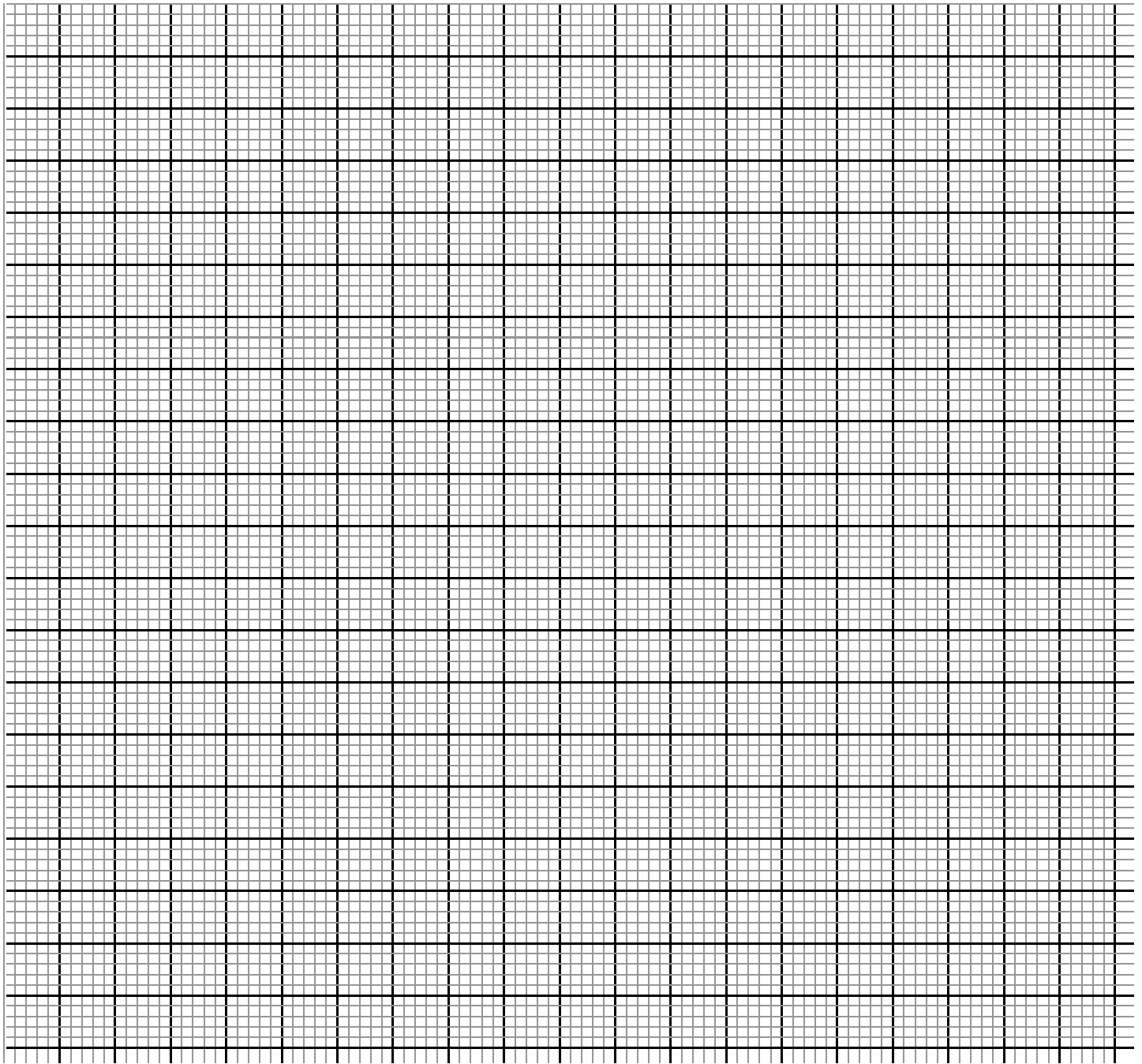
d) The maximum velocity attained **(2 marks)**

23. Complete the table below for $y = \sin 2x$ and $y = \sin (2x + 30)$ giving values to 2d.p

x°	0	15	30	45	60	75	90	105	120	135	150	165	180
$\sin 2x$	0				0.87				-0.87				0
$\sin (2x + 30)$	0.5				0.5				-1				0.5

(2 marks)

b) Draw the graphs of $y = \sin 2x$ and $y = \sin (2x + 30)$ on the axis. **(5 marks)**



c) Use the graph to solve $\sin (2x + 30) - \sin 2x = 0$

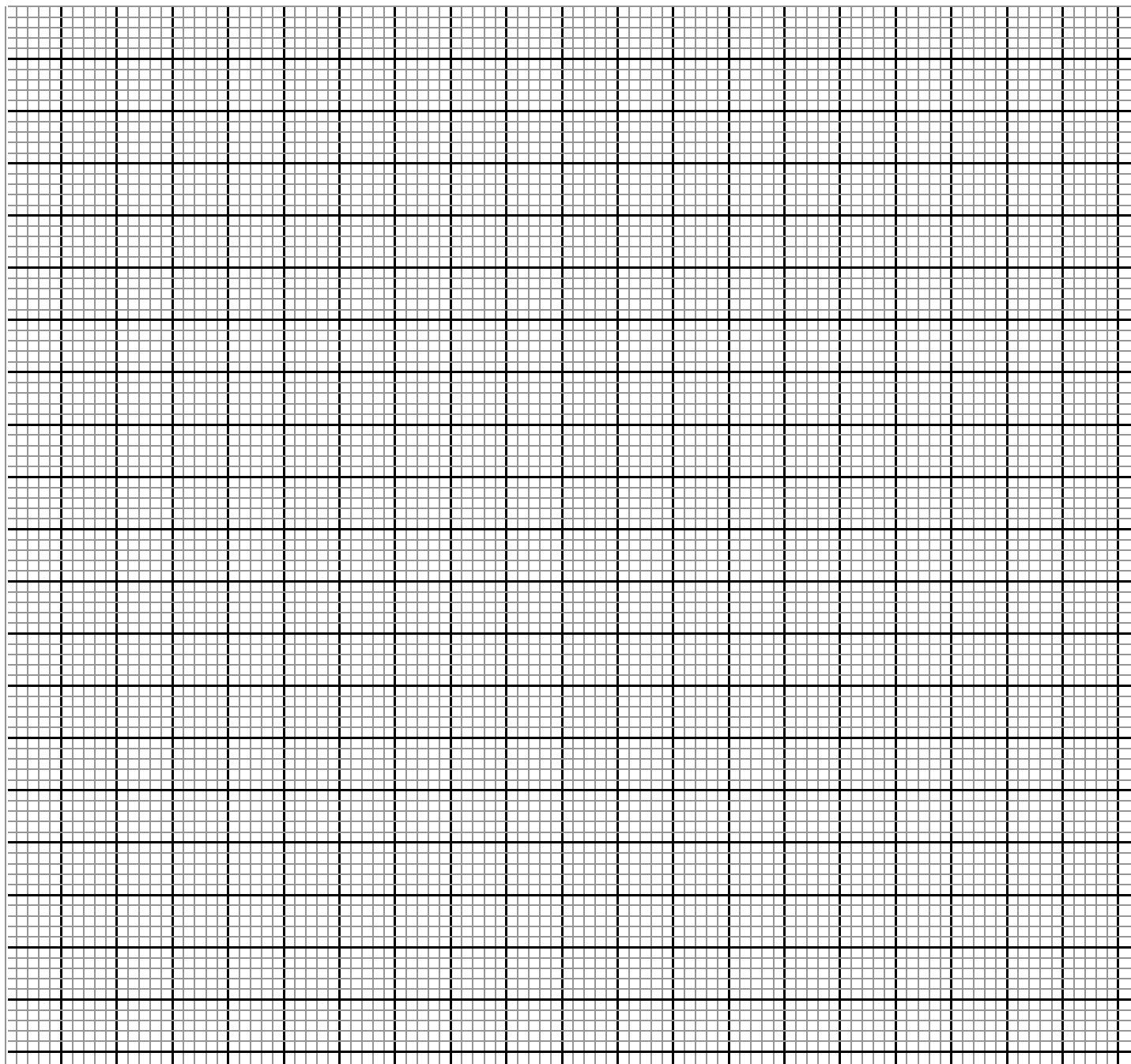
(1 mark)

- d)** State the period and amplitude of $y = \sin (2x + 30)$
(2marks)

24. A trader makes chairs and tables. The cost of each chair is shs.300 while each table costs shs.700. He is prepared to spend not more than shs. 21,000. It is not viable for him to make less than 20 items. chairs must be less than twice the tables but more than 15. By taking the number of chairs as x and tables as y :
- (a) Write down all the inequalities in x and y . (4marks)

- (b) Draw the inequalities on the grid provided.

(4marks)



(c) He sells tables at a profit of shs.140 while chairs at a profit of shs.120;

Determine the maximum possible profit he makes.

(2 marks)

“END”